Maya Backend: CTO-Level Technical Strategy for an AI-Native CLO3D Replacement

TL;DR

- **The fastest viable path to a CLO3D replacement today is a hybrid stack**: fine-tune a multimodal foundation model in the AIpparel / ChatGarment / Design2GarmentCode family to predict GarmentCode (Korosteleva/Sorkine-Hornung, ETH Zurich) sewing-pattern programs from images, drape with a modified NVIDIA Warp simulator (the PyGarment fork), and persist GarmentCode JSON as the canonical 2D↔3D representation. Build the user-facing 2D/3D linked panels as your own UI on top — but keep CLO3D in the loop as an export/QA target via its Python plug-in API while Maya matures.
- **The research field reached production-relevance in 2024–2025**: GarmentCodeData (115K 3D garments, ECCV 2024), ChatGarment / AIpparel / Design2GarmentCode (all CVPR 2025), GarmentDiffusion and GarmageNet (SIGGRAPH Asia 2025, with code from Style3D), and Dress-1-to-3 (TOG 2025) converge on the same pipeline: image → VLM/diffusion → parametric pattern program → differentiable drape → 3D mesh. korosteleva
- **Build vs. buy/partner**: Build the pattern-generation core in-house (it is a defensible IP wedge and the open-source state of the art is good enough to fine-tune on your own production data). Buy/license: NVIDIA Warp + (optionally) Marvelous Designer SDK as a fallback drape engine. Partner: nobody yet — the commercial leaders (Style3D, CLO Virtual Fashion, Browzwear) are all building this stack themselves and won't license. Watch Refabric and AI.Fashion as potential acquisition targets if you need talent.

Key Findings

1. The image→pattern→3D research landscape has consolidated around GarmentCode

The dominant 2D pattern representation across virtually every 2024–2026 paper is now GarmentCode (Korosteleva & Sorkine-Hornung, SIGGRAPH Asia 2023, MIT-licensed) and its dataset GarmentCodeData (115K garments × 5K body shapes, ECCV 2024). It is a parametric Python program (with a JSON serialization) that encodes panels, edges, stitches, and 3D placement. AIpparel, ChatGarment, Design2GarmentCode, GarmageNet, and GarmentDiffusion all use it (or close variants) as their target representation. This convergence is highly favorable for a builder: you can pick from ~6 production-quality models that all consume/emit the same format. Maria Korosteleva korosteleva

2. Three publicly viable end-to-end image-to-pattern pipelines exist today

System	Conference	Input	Pipeline	Code
ChatGarment (Bian et al., MPI/SJTU)	CVPR 2025	image or text	Fine-tuned VLM → GarmentCode JSON → drape Chatgarment	chatgarment.github.io
AIpparel (Nakayama, Ackermann, Kesdogan et al., Stanford/ETH)	CVPR 2025 (highlight)	image, text, sketch	Fine-tuned LMM with custom sewing-pattern tokenizer Georgenakayama ResearchGate → GarmentCodeData JSON	github.com/georgeNakayama/AI Code + HF dataset
Design2GarmentCode (Zhou, Liu, Wang et al., Style3D Research + Zhejiang)	CVPR 2025	image, sketch, text	LMM → GarmentCode **Python program** (not JSON) → execution engine	github.com/Style3D (announcec partial release)
Dress-1-to-3 (Li, Yu, Du, Jiang et al., UCLA)	TOG 2025	single image	Image → sewing pattern (Sewformer) + multi-view diffusion → differentiable Warp drape → simulation-ready 3D Aiartweekly arXiv	dress-1-to-3.github.io
GarmageNet (Style3D + SJTU + Zhejiang)	SIGGRAPH Asia 2025	image, text	Latent-diffusion transformer over "Garmage" geometry-image rep → patterns + 3D init + stitching (GarmageJigsaw	github.com/Style3D/garmagene

System	Conference	Input	Pipeline	Code
			integrates into Style3D Studio)	
GarmentDiffusion (Style3D + Shenfu Research)	2025	multimodal	Edge-token diffusion transformer over sewing patterns arXiv GitHub	github.com/Shenfu-Research/GarmentDiffusion
SewFormer (Sea AI Lab / Liu, Xu, Lin, Liang, Yan)	SIGGRAPH Asia 2023	single image	Two-level transformer → discrete sewing pattern Sewformer	github.com/sail-sg/sewformer
Panelformer (WACV 2024)	WACV 2024	2D image	Panel + stitch transformer Ericsujw	ericsujw.github.io/Panelformer

The pragmatic stack today: **ChatGarment or AIpparel for the image→GarmentCode model, plus the PyGarment + NVIDIA Warp fork (maria-korosteleva/NvidiaWarp-GarmentCode) for the simulator.** Both are openly available; both have working released code; both target the same GarmentCode JSON schema.

3. The "draping" half is solved by NVIDIA Warp + PyGarment v2.0

Korosteleva's lab released `NvidiaWarp-GarmentCode` (a fork of NVIDIA Warp v1.0.0-beta.6 with the cloth-simulation modifications used in the GarmentCodeData paper). PyGarment v2.0.0 (Aug 2024) wires this together end-to-end with mesh generation, simulation, GUI, and sampling. NVIDIA Warp itself (~5.4k GitHub stars) is GPU-accelerated, differentiable, JIT-compiled — production-grade. The base Warp's commercial license was expanded by NVIDIA to permit commercial use; the Korosteleva fork follows that licensing decision (per the repo README). This means the drape engine is essentially free for Mana Siyo, no royalty obligation. GitHub + 5

Alternative differentiable simulators that may matter:

- **DiffCloth** (Yifei Li et al., MIT CSAIL, TOG 2022/SIGGRAPH 2023) — Projective-Dynamics-based with dry frictional contact. C++/CUDA, Python bindings. Source: github.com/omegaiota/DiffCloth. Used in Dress-1-to-3 for sewing-pattern refinement. ACM Digital Library MIT CSAIL

- **C-IPC** (Codimensional Incremental Potential Contact, Li, Kaufman, Jiang) — most physically accurate but slow.
- **ARCSim** — Classic; deprecated for new work but still cited.

4. The image→3D-clothed-human research lineage (everything ChatGarment / Dress-1-to-3 builds on)

Critical lineage:

- **SMPL / SMPL-X / SMPL-H** (Loper, Romero, Pons-Moll, Black, MPI) — the universal parametric body model.
- **Multi-Garment Net (MGN)** (Bhatnagar, Tiwari, Theobalt, Pons-Moll, ICCV 2019).
- **SMPLicit** (Corona, Pumarola, Alenya, Pons-Moll, Moreno-Noguer, CVPR 2021).
- **BCNet** (Jiang et al., ECCV 2020).
- **DeepFashion3D / DeepFashion3D V2** (Zhu, Cao, Jin, Han et al., CUHK-SZ, ECCV 2020) — 2,078 reconstructed garment models across 10 categories. arxiv
- **PIFu / PIFuHD / ICON / ECON** (Saito, Black, Xiu et al., USC/MPI) — pixel-aligned implicit functions for clothed human reconstruction; ECON is the current SOTA for in-the-wild single-image clothed humans.
- **ReEF** (Zhu et al., 2022) — Registering Explicit to Implicit.
- **ClothCap** (Pons-Moll, Pujades, Hu, Black, SIGGRAPH 2017) — original 4D capture work.
- **TailorNet** (Patel, Liao, Pons-Moll, CVPR 2020).
- **DrapeNet** (De Luigi, Li, Guillard, Salzmann, Fua, CVPR 2023) — EPFL/CVLab.
- **ISP: Implicit Sewing Patterns** (Li, Guillard, Fua, NeurIPS 2023) — github.com/liren2515/ISP. The closest academic precursor to fitting sewing patterns to images via gradient descent. Extended to **DISP** (Diffusion-based) for spatio-temporal video reconstruction (2025).
- **GarVerseLOD** (2024) — large LOD-structured 3D garment dataset for high-fidelity reconstruction.
- **GarmentDreamer** (Li et al., 3DV 2025) and **ClotheDreamer** (Liu et al., 2024) — 3D Gaussian Splatting–guided text-to-garment generators that produce simulation-ready meshes. Text-driven, not image-driven, and do NOT produce sewing patterns. OpenReview
- **DressWild** (2026 preprint) — feed-forward pose-agnostic garment sewing-pattern generation from in-the-wild images. Latest frontier. ResearchGate ResearchGate
- **GarmentX** (Guo et al., arXiv 2504.20409, 2025) — autoregressive parametric pattern representation. ACM Digital Library
- **GarmentGPT** (ICLR 2026, Weng et al., CUHK-SZ) — discrete latent tokenization for compositional pattern generation. github

5. Commercial competitive landscape — what is publicly known

Company	HQ	Founded	Total raised	Most recent priced round	Product focus
CLO Virtual Fashion (CLO3D, Marvelous Designer)	Seoul	2009	~$70M+	**KRW 50B (~$34.8M USD), Dec 16 2024**, Atinum Investment + IMM Investment The Pickool (KED Global reports $35M; WOWTALE $36.1M — variance reflects exchange-rate timing)	3D CAD apparel · VFX/games. KoreaTechDesk C shareholding wit Games. 3DVF
Style3D / Linctex Digital	Hangzhou	2015	~$119–145M	**~$100M Pre-Series B+, June 2022**, Hillhouse (GL Ventures) + CDH PYMNTS.com Crunchbase	Studio 3D CAD + Atelier (AI) + GarmageNet rese engine; expansio EU/US since 2021 Style3D Crunch Acquired Assyst . 2023. CB Insights
Browzwear	Singapore (R&D Israel)	1999	$35M	**$35M growth equity, Aug 2021**, Radian Capital Pangaia	VStitcher 3D CAI Crunchbase + Lo **acquired Lalalan on July 25 2025** Yahoo Finance pe FashionUnited ("Browzwear, a le in digital product creation (DPC) fo fashion, has annc its full acquisitior Lalaland.ai");

Company	HQ	Founded	Total raised	Most recent priced round	Product focus
					fashionunited The Next Web ne CEO Greg Hanso 2024. Crunchbase
SEDDI	Madrid + NYC	2018	grants + small inst.	n/a	Author 3D CAD + Textura fabric digitization; **13 pa filed**; Crunchbase + Spanish CDTI funded; URJC sp (Otaduy).
3DLOOK	San Mateo (ex-Ukraine)	2016	~$11.2M Retail News USA	$3.5M Series A extension, Jan 2022 Tracxn	2-photo body sca measurements + No 2023+ fundin capital-constrain
Bold Metrics	San Francisco	2012	~$12.3M	**$8M Series A, May 2022**, Bessemer Venture Partners Yarnsandfibers	AI body sizing fr questions. Boldm Per Built In SF's coverage of the S A: "Companies u Bold Metrics — including Canad Goose, Tailored B (Men's Wearhou A. Bank), SuitSh Blue Delta Jeans, UpWest and SIM Built In San Francis
The Fabricant	Amsterdam	2018	$14M+	$14M Series A, March 2022, Greenfield One (with Sound Ventures, Red DAO) Silicon Canals	Digital-only cout NFTs; momentu cooled with NFT market.

Company	HQ	Founded	Total raised	Most recent priced round	Product focus
Reactive Reality	Graz	2014	undisclosed	accelerator only (Plug and Play) Crunchbase Crunchbase	PICTOFiT virtual on; Crunchbase **patents filed**; Crunchbase Hu Boss customer. Crunchbase
Resleeve.ai	Eindhoven	2021	~€3M	Seed €3M, 2024 (undisclosed lead) SpeakerHub	AI photoshoot / C to-render Yahoo Finance + try-on. Wellfound **YC**, not pattern generation.
Refabric (Mintgrams)	Delaware/Istanbul/Paris	2022	undisclosed	Sun Tekstil strategic investment, Feb 2025 Textileworld	Gen-AI fashion d Crunchbase clai **patternmaking + waste pattern optimization**. World-collective La Maison des Startups. Crunch Closest public competitor in ima pattern AI.
AI.Fashion	Los Angeles (YC)	2023	$3.6M	Seed, Feb 2024, led by Neo FashionUnited	YC's bet on the sp image generation/photo not patterns.
CALA	NYC	2016	~$5.5M	$3M seed, July 2020, Maersk Growth + Real Ventures Tracxn FinSMEs	"OS for fashion" design+supply ch Crunchbase no f capital in 5 years.
Unmade	London	2013	**£15M total** (per Unmade's July	**Acquired by Hi-Tech**	UnmadeOS for knitwear/custom

Company	HQ	Founded	Total raised	Most recent priced round	Product focus
			9 2024 press release: "the award-winning tech company had raised a combined £15 million from Octopus Ventures, Felix Capital, MMC Ventures, Local Globe and Connect Ventures") Unmade UK Tech News	**Apparel, July 2024** Unmade +3	mfg.

Consolidation signal (very important for strategy): Browzwear → Lalaland.ai (July 25, 2025), Style3D → Assyst (Jan 2023), Hi-Tech Apparel → Unmade (July 2024), House of Blueberry → The Fabricant (signal, unconfirmed). The market is contracting into a **3-player race (Style3D, CLO Virtual Fashion, Browzwear)** for the integrated 3D-CAD + AI design suite, with smaller players being absorbed or starved. fashionunited Unmade

6. Patents and IP landscape — moderate risk, not blocking

- **CLO Virtual Fashion** has issued patents including US10733773B2 ("Method and apparatus for creating digital clothing"), US11410355B2, and patents on dart generation (Lee, Kim, Pinkham), 3D draping simulation with 2D pattern input, and constrained-area garment simulation. These are largely about UI and simulation method specifics; nothing in the patent set obviously blocks an AI-first pipeline that learns to predict GarmentCode programs and renders via Warp. Justia Patents
- **Style3D (Linctex)** has filed patents on its proprietary soft-tissue simulation engine in China; less visible in USPTO.
- **SEDDI** holds 13 patents, mostly around yarn-level fabric digitization and simulation. Of concern if Maya touches fabric digitization from photos. Crunchbase
- **Recurring USPTO patent (Garment pattern generation from image data, US 12,198,290)** explicitly claims a system that ingests image data, produces 3D mesh + flat sewing pattern panels, virtually drapes on a 3D body, and renders the dressed image. The claim language is broad. **Assignee and freedom-to-operate analysis on this patent should be the single**

highest-priority external-counsel item before any commercial launch. Source: image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/12198290. uspto

- **Knitwear**: Shima Seiki (Japan) holds >2,900 patents on WHOLEGARMENT® knitting tech — not relevant unless Mana Siyo extends into seamless knit. PR Newswire
- **Recommendation**: Engage a patent attorney specifically on US 12,198,290 and its assignee before any external launch. Published image-to-pattern academic work (NeuralTailor 2022, SewFormer 2023, GarmentCode 2023) likely constitutes substantial prior art against overly broad claims, but only counsel can confirm.

7. CLO3D's developer surface area — usable but not a replacement

CLO ships a real SDK at developer.clo3d.com. Key facts:

- C++ API (.dll/.dylib) since v2.x, **Python plug-in support added recently** (.py files registerable via Plug-in Manager). Clo3d
- Library Window Plug-in API lets you replace the asset/PLM panel with your own data source. Clo3d
- Event plug-in lets you intercept drag-drop of fabrics/files into 2D or 3D windows. Clo3d
- Python API allows scripting from inside CLO via Main Menu → Edit Python Script. Clo3d
- The API exposes Pattern, Fabric, Utility, and other namespaces; you can collect BOM, manipulate patterns programmatically. Clo3D Support
- **What it does NOT expose**: training data, the simulation kernel, or the 2D↔3D linkage internals (closed and patent-protected).

Maya can interoperate with CLO (import/export patterns, push generated patterns into CLO for QA and laser-cutter export, run scripts via Python) but **cannot reuse CLO's simulation engine** to power Maya itself. The right tactic is to keep CLO as a verification target during the transition (Maya generates → CLO loads → laser cut), then phase it out as Maya's own 3D renderer matures.

8. Datasets available — large enough to train

- **GarmentCodeData v2** (Korosteleva et al., ECCV 2024): 115K 3D garments × 5K body shapes. ETH Research Collection doi 10.3929/ethz-b-000690432. Best primary training set today. korosteleva
- **GarmentCodeData-Multimodal** (Nakayama et al., CVPR 2025): GCD extended with images and text annotations — the dataset behind AIpparel. HF: huggingface.co/georgeNakayama/AIpparel.
- **SewFactory** (Liu et al., 2023): ~1M images with sewing patterns + 3D pose + densepose. Released with Sewformer. Sewformer
- **GarmageSet** (Style3D, Nov 2025): GarmageNet's dataset, CC BY-NC-ND 4.0 (research only).

- **DressCodeData** (He et al., 2024): pattern + text-aligned dataset behind DressCode.
- **DeepFashion3D / V2** (CUHK-SZ): 2,078 reconstructed garments + feature lines + pose + multi-view real images. Form-gated download.
- **4D-DRESS / Dress4D / CloSe**: 4D scans of people in clothing for dynamic reconstruction. Mostly research-licensed.
- **CLOTH3D** (Bertiche, Madadi, Escalera): synthetic 3D garments on SMPL. Research license.
- **DeepFashion-MultiModal**: rich text annotations on the original DeepFashion image set.

Commercial-use licensing reality check: GarmentCode itself is MIT-licensed, but GarmentCodeData v2 is CC-licensed (ETH Research Collection — verify exact CC variant before commercial training). GarmageSet is explicitly CC BY-NC-ND 4.0 (no commercial use, no derivatives). SewFactory's exact terms must be checked. **Mana Siyo should also build a proprietary dataset from its own production garments (24-hour bespoke shop produces a rich stream of paired image + pattern + finished-garment data — this is the real moat).**

9. Foundation model and segmentation infrastructure

- **SAM 2** (Meta, 2024) — best segmentation for separating garments from bodies/backgrounds. Apache 2.0.
- **DINOv2** (Meta) — best self-supervised image features for fashion understanding; Apache 2.0. Useful as a frozen backbone.
- **Qwen2-VL, LLaVA, InternVL** — open-source VLMs suitable for fine-tuning into the AIpparel/ChatGarment role. ChatGarment used LISA-style fine-tuning over a Vicuna+CLIP base; AIpparel used a similar LMM backbone with a custom pattern tokenizer.
- **Multi-view diffusion priors** for the image→multi-view step: SV3D, Zero123++, MVDream, Era3D, Wonder3D — Dress-1-to-3 specifically uses a pre-trained multi-view diffusion model for refining the pattern. arXiv The Moonlight
- **3D Gaussian Splatting** (Kerbl et al., SIGGRAPH 2023) — used in GarmentDreamer and ClotheDreamer for text-to-garment generation, but not a strong fit for the pattern-output workflow because GS produces splat clouds that need mesh extraction. Useful as a visualization layer if Maya wants a high-fidelity render.

10. Mesh-to-pattern / unwrapping libraries (the "drape→flatten" half)

If Maya ever needs to go from a 3D mesh back to a 2D pattern (e.g., user edits a 3D drape and wants the pattern updated):

- **BFF (Boundary First Flattening)** — Sawhney & Crane, SIGGRAPH 2018. C++ + Python bindings. github.com/GeometryCollective/boundary-first-flattening. Best modern algorithm for low-distortion UV flattening.

- **LSCM (Least Squares Conformal Maps)** — Lévy et al. Available in libigl, OpenMesh, Geometry Central.
- **ABF / ABF++ (Angle-Based Flattening)** — Sheffer et al. In CGAL and libigl.
- **libigl** (Jacobson, Panozzo) — C++/Python; covers LSCM, ARAP parameterization, mesh slicing, planar parameterization.
- **Geometry Central** (Sharp, Soliman, Crane) — modern C++ library; integrates BFF, intrinsic triangulations.

All open-source and well-maintained.

Details

Recommended technical architecture for Maya (concrete stack)

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                 ┌───────────────────────────────────────┐
                 │  USER: customer photo or designer brief │
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                               ▼
          ┌─────────────────────────────────────────┐
          │  STAGE 1 — Perception (CPU/light GPU)    │
          │  • SAM 2 — segment garment(s)            │
          │  • DINOv2 features                       │
          │  • Optional: 4D-Humans / SMPL-X fit for body │
          │  • Optional: ECON for clothed-human geometry │
          └────────────────────┬────────────────────┘
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                               ▼
          ┌─────────────────────────────────────────┐
          │  STAGE 2 — Pattern Synthesis (A100/H100) │
          │  • Fine-tuned Qwen2-VL or LLaVA backbone │
          │  • AIpparel-style tokenizer for GarmentCode │
          │  • Output: GarmentCode JSON (panels + edges + │
          │    stitches + 3D placement)              │
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          ┌─────────────────────────────────────────┐
          │  STAGE 3 — Draping (single GPU, 5-60s)   │
          │  • PyGarment v2 mesh generation          │
          │  • NvidiaWarp-GarmentCode simulator      │
          │  • Output: 3D garment mesh on SMPL body  │
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          ┌─────────────────────────────────────────┐
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       │  STAGE 4 — Refinement (optional)                   │
       │  • Multi-view diffusion (Zero123++/SV3D)           │
       │  • Differentiable Warp drape vs. rendered views    │
       │  • Backprop into GarmentCode params (Dress-1-      │
       │    to-3 style)                                     │
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       ┌────────────────────────────────────────────────────┐
       │  STAGE 5 — Manufacturing post-processing           │
       │  • Seam allowance offsetting (per edge label)      │
       │  • Grainline assignment (per panel)                │
       │  • Notches (from seam pairings)                    │
       │  • Layout/nesting for laser cutter                 │
       │    (deepnest.io or commercial)                     │
       │  • DXF/AAMA/SVG export                             │
       └────────────────────────────────────────────────────┘

UI layer: linked 2D-panel ↔ 3D-mesh viewer (web: three.js + a custom 2D
SVG/Canvas pattern editor; or desktop: PySide6 + VTK / Open3D).
GarmentCode JSON is your single source of truth — both views are
projections of it.
```

The key architectural insight: **GarmentCode JSON is the document.** The 2D panel editor and the 3D drape are both *views* of the same document, not independent objects. This mirrors CLO3D's internal linkage (patent-protected as a UI but trivially reimplementable on top of a parametric pattern program).

GPU and latency estimates (realistic, not vendor-claimed)

Stage	Hardware	Latency (1 garment)	Notes
SAM 2 + DINOv2 + SMPL fit	1× consumer GPU (RTX 4090)	2–5 s	Real-time
AIpparel/ChatGarment inference	1× A100 80GB or 2× 4090	5–15 s	Larger LMM = better accuracy, slower
GarmentCode → mesh	CPU	<1 s	Pure geometry
Warp drape (XPBD, T-pose, 60 frames)	1× A100	15–60 s	Scales with mesh resolution

Stage	Hardware	Latency (1 garment)	Notes
Multi-view diff refinement (optional)	1× A100	60–180 s	Only for in-the-wild images
Total (no refinement)	A100	**~30–80 s**	Realistic
Total (with refinement, Dress-1-to-3 style)	A100	**3–5 min**	High-fidelity

Cloud A100 80GB pricing varies materially: per Spheron's May 2026 GPU pricing comparison, "Cheapest A100 80GB per hour: Spheron at $1.07/hr on-demand and $0.60/hr spot." RunPod Community Cloud A100 80GB starts at ~$1.39/hr, and Lambda Labs lists A100 80GB at $2.49/hr on-demand. Budget $1.50–$2.50/hr on mainstream providers; cheaper if you can use Spheron-class spot capacity. Even at $2.49/hr, each pattern costs single-digit cents in compute. Well within budget for a 24-hour bespoke workflow. Spheron SynpixCloud

Build/buy/partner decision matrix

Component	Recommendation	Rationale
Image→pattern model	**Build** (fine-tune AIpparel/ChatGarment)	Defensible IP; your production garments are unique training data
2D pattern representation	**Adopt GarmentCode (MIT)**	Industry-converging standard; saves 12+ months of representation design
Drape simulator	**Adopt NvidiaWarp-GarmentCode**	Free, GPU-accelerated, differentiable, commercial-permitted
Body model	**Adopt SMPL-X** (commercial license)	Universal; license is NOT standard open-source — see caveats
Mesh flattening (if needed)	**Adopt BFF + libigl**	Best-in-class open source
Segmentation/perception	**Adopt SAM 2 + DINOv2**	Apache 2.0
2D/3D linked viewer	**Build**	UI is the customer experience; do not depend on third party
Laser-cutter nesting & DXF export	**Build on deepnest.io or buy**	Many open + commercial options; not a research problem

Component	Recommendation	Rationale
CLO3D interop	**Partner / use SDK**	Keep CLO as QA target during transition; deprecate after Maya is verified
Patent freedom-to-operate	**Buy external counsel**	Specifically on US 12,198,290 and CLO Virtual Fashion patents

Phased implementation roadmap

Phase 0 — Foundations (weeks 1–4)

- Lock in GarmentCode JSON as the canonical representation.
- Stand up an internal dataset pipeline: every garment Mana Siyo produces gets photographed (multi-view), its CLO3D pattern exported to GarmentCode JSON, its finished form re-photographed. Target: 500 paired examples in 30 days.
- Reproduce ChatGarment or AIpparel inference end-to-end on a few sample images. Establish baseline accuracy.

Phase 1 — Pattern model prototype (weeks 5–16)

- Fine-tune AIpparel-style model on GarmentCodeData-Multimodal + your initial dataset.
- Build the linked 2D/3D viewer (web app using three.js for 3D and a custom SVG editor for 2D; both reading/writing the same GarmentCode JSON store).
- Hook up the NvidiaWarp-GarmentCode drape and show 2D↔3D in a single screen.

Phase 2 — Production-quality output (weeks 17–32)

- Implement seam-allowance offsetting, notches, grainline as deterministic post-processing on the GarmentCode JSON.
- Implement DXF/AAMA-DXF export validated against your laser cutter.
- Integrate Dress-1-to-3-style multi-view refinement for hard in-the-wild images.
- Begin A/B testing against current CLO3D workflow: time-to-pattern, accuracy of fit, customer satisfaction.

Phase 3 — CLO3D deprecation and external launch (months 9+)

- Once Maya's output passes the same QA as the current CLO workflow for >95% of garments, transition the laser-cutter path to Maya-only.
- External counsel patent FTO before any external launch.
- Productize as either internal-only or as a SaaS for other bespoke houses.

Open questions and unknowns

1. **Exact license of GarmentCodeData v2** — ETH Research Collection lists it under a CC license but the specific variant (BY vs. BY-NC) materially affects whether you can train a commercial model on it. Verify directly with ETH IGL.
2. **Assignee and claim scope of USPTO 12,198,290 (garment pattern generation from image data)** — single highest IP risk; must be cleared by counsel.
3. **Whether Style3D will release Design2GarmentCode code commercially** or keep it inside Style3D Studio. GarmageNet is explicitly CC BY-NC-ND — Design2GarmentCode may follow the same pattern, in which case you cannot use it as a base.
4. **In-the-wild robustness gap.** Every published image-to-pattern model is trained primarily on rendered/clean images. Real customer photos have lighting variation, occlusion, complex poses. Plan for a significant domain-adaptation effort with your own data. ResearchGate
5. **Fabric properties from photo.** None of the published image→pattern systems estimate fabric drape parameters from the image. Either use a fabric-library lookup (you know what fabrics you stock) or follow SEDDI's research direction (which is patent-protected).
6. **Whether CLO Virtual Fashion's $34.8M Dec 2024 round funds an AI-pattern-from-image product**. Atinum and IMM are growth-equity, not deep-tech investors; signals that CLO is going for distribution, not R&D leadership. Good for you — the incumbent is unlikely to leapfrog the open research. MarketScreener
7. **Refabric's exact claims about "AI patternmaking"**. They claim it; no published method. Worth a discovery call or trade-show conversation to assess. World-collective

Recommendations

1. **Commit to GarmentCode JSON as Maya's canonical pattern representation.** Adopt PyGarment v2 + NvidiaWarp-GarmentCode as the drape engine. This is a 12-month head start over building any of this in-house.
2. **Fine-tune AIpparel (Stanford/ETH) as the image→pattern model** — cleanest released code, best dataset (GarmentCodeData-Multimodal), strongest author team (Korosteleva, Wetzstein, Sorkine-Hornung, Guibas). Avoid GarmageNet as a base because of its CC BY-NC-ND license. Use ChatGarment as a second comparison baseline.
3. **Immediately start photographing every garment in production.** Within 6 months Mana Siyo's proprietary paired-data corpus will be the single most defensible asset in this stack. The bespoke 24-hour workflow is the moat — the AI is commoditizing fast.
4. **Engage IP counsel on USPTO 12,198,290 and CLO patents before any external launch**, even before fundraising. The cost of an FTO opinion is rounding error relative to a downstream injunction.

5. **Keep CLO3D in the loop via its Python plug-in API** as the QA/export target until Maya can verifiably match its output. Build a script that round-trips GarmentCode JSON ↔ CLO project files.
6. **Hardware: start on a single A100 80GB** in the cloud ($1.07–$2.49/hr depending on provider — see Spheron/RunPod/Lambda pricing in §Details) for model training and inference. For local production, a workstation with one or two RTX 4090s (24GB each) will run inference in real time and is far cheaper than an on-prem H100.
7. **Hire one researcher with sewing-pattern-graph experience.** The field is small enough that 10–15 people in the world have deeply published here. Candidates to track: alumni of Korosteleva's lab (ETH IGL), the Style3D Research team (Huamin Wang, Ruiyang Liu, Siran Li), Sea AI Lab pattern team (Lijuan Liu, Xiangyu Xu), Pascal Fua's group at EPFL (Ren Li, Benoît Guillard). Korosteleva herself is at Epic Games now and likely off the market. Maria Korosteleva korosteleva
8. **Watch but do not partner with Refabric, Style3D, Browzwear, or CLO.** They are competitors. The space is consolidating; a tactical acquisition signal from Mana Siyo to a small player (e.g., Refabric, AI.Fashion) may be more useful than a license deal.

Benchmarks that would change these recommendations

- If GarmentCodeData v2's license confirms BY-NC: pivot to building a proprietary dataset before scaling training.
- If USPTO 12,198,290 has a hostile assignee and broad enforceable claims: pivot to a structurally different output representation (e.g., pure 3D mesh with on-device flattening, no explicit "drape on body" step) or seek a license.
- If Style3D releases Design2GarmentCode under MIT/Apache: re-evaluate using it as a stronger base than AIpparel for sketch-conditioned generation.
- If Refabric raises a meaningful (>$10M) Series A in 2026: assume image-to-pattern is becoming a contested commercial category and accelerate launch.

Caveats

- Funding figures for non-US companies (Style3D, Refabric, Resleeve, Reactive Reality) rely on Chinese, Turkish, and Dutch press releases and may be partial. Resleeve's €3M is from a founder-supplied bio rather than a press release. The Fabricant's possible absorption by House of Blueberry is from a Crunchbase signal and is not independently verified.
- Several of the most exciting 2025–2026 papers (DressWild, GarmentGPT-ICLR 2026, Dress-1-to-3) had not released full code at the time of writing; code-availability claims are based on what authors had on project pages and GitHub orgs as of mid-2026.
- Patent claim interpretation in this report is not legal advice. The discussion of USPTO 12,198,290 is based on the published abstract and claim 1 only and requires a formal FTO opinion.

- Latency estimates assume an A100 80GB. Consumer hardware (RTX 4090) will be 1.5–2.5× slower per stage; H100 ~1.3–1.5× faster than A100 but ~3× cost per hour in the cloud.
- The "GarmentCode is the document" architectural recommendation depends on GarmentCode's continued maintenance. Korosteleva moved from ETH to Meshcapade and then to Epic Games. If ETH IGL or Epic deprioritize the project, Mana Siyo would inherit maintenance — budget for one engineer's time to fork and maintain it. Google Scholar + 2
- No commercial-use license review of SMPL/SMPL-X has been done here. MPI's license is NOT standard open-source; commercial deployment requires a separate agreement with Max Planck or with Meshcapade. Use SMPL-X under an explicit commercial license; do not assume open use.
- Bold Metrics customer list excludes Levi's despite earlier sources mentioning it — Levi's partnered with Lalaland.ai (now owned by Browzwear) in 2023, not with Bold Metrics. Confirmed Bold Metrics customers per Built In SF: Canada Goose, Tailored Brands (Men's Wearhouse, Jos A. Bank), SuitShop, Blue Delta Jeans, UpWest, SIMMS. Just Style + 2